WASP-2b

R-1287

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-2_210909` · created 2026-07-31T15:46:15+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459464.6274 +/- 0.006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.037
Transit depth (Rp/Rs)^2	2.55 +/- 1.18 [%]
Orbital Inclination (inc)	89.55 +/- 2.08
Airmass coefficient 1 (a1)	0.8472 +/- 0.0067
Airmass coefficient 2 (a2)	0.15 +/- 0.13
Scatter in the residuals of the lightcurve fit is	0.8 %
Best Comparison Star	#2 - [430, 240]
Optimal Aperture	3.94
Optimal Annulus	11.8
Transit Duration (day)	0.0367 +/- 0.0068

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.16 ± 0.037 vs 0.1326 (deviation 0.74σ)
Duration fit vs published	0.0367 d vs 0.0737 d (deviation 5.45σ)
Mid-time O–C	-3082.6 min
Depth SNR	4.3
Residual scatter	0.8 %

Light curve

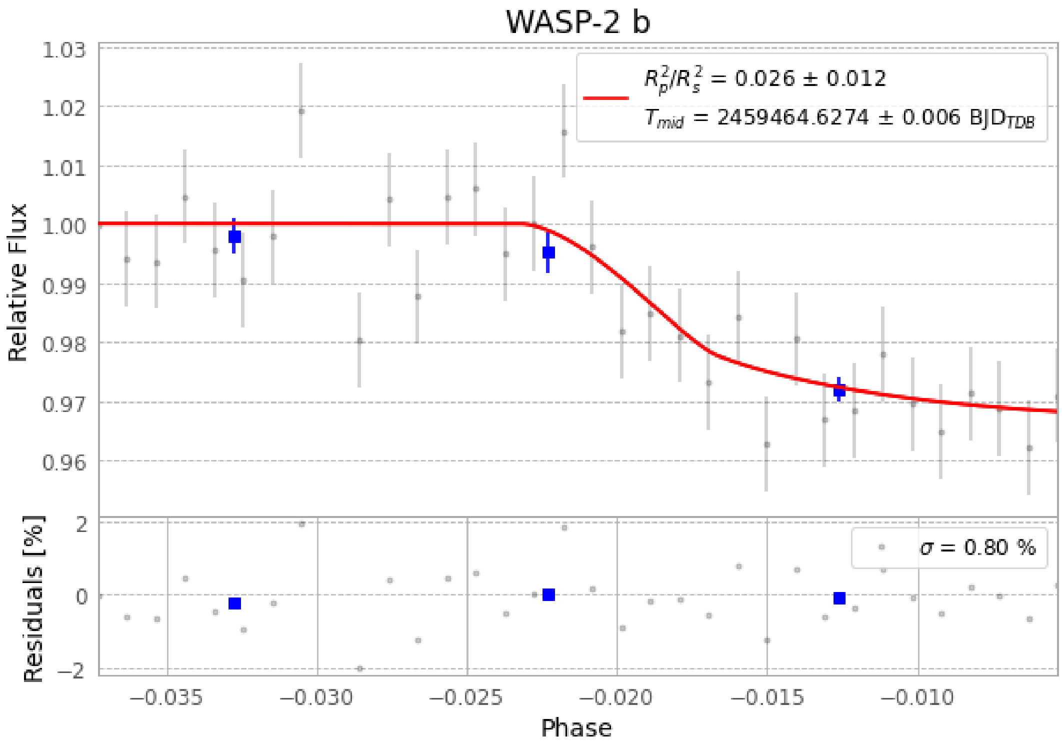

Figure 1 — FinalLightCurve_WASP-2 b_2021-09-08.png (SHA-256 ab94f0160d2cdffe…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [388, 124], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 3ad9e70626df9511…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

34 FITS frames (22 MB), `WASP-2210909043912.FITS` → `WASP-2210909061812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-210909.log (c33df3570c31…), FinalLightCurve_WASP-2 b_2021-09-08.png (ab94f0160d2c…), FinalLightCurve_WASP-2 b_2021-09-08.csv (4fb054aeb524…)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)